



CLO 1.5

Describe Data Using Measures of Variation such as the Range, Variance and Standard Deviation (population and sample).

BUS 1023: Applied Mathematics for Business

Monday, September 21, 2026

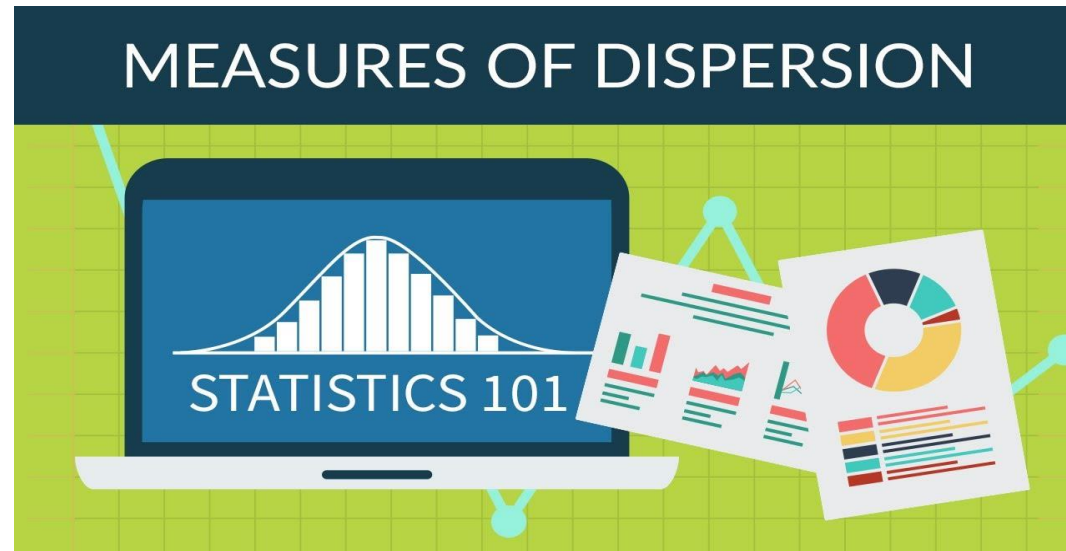


Measures of Dispersion (Variations)

2

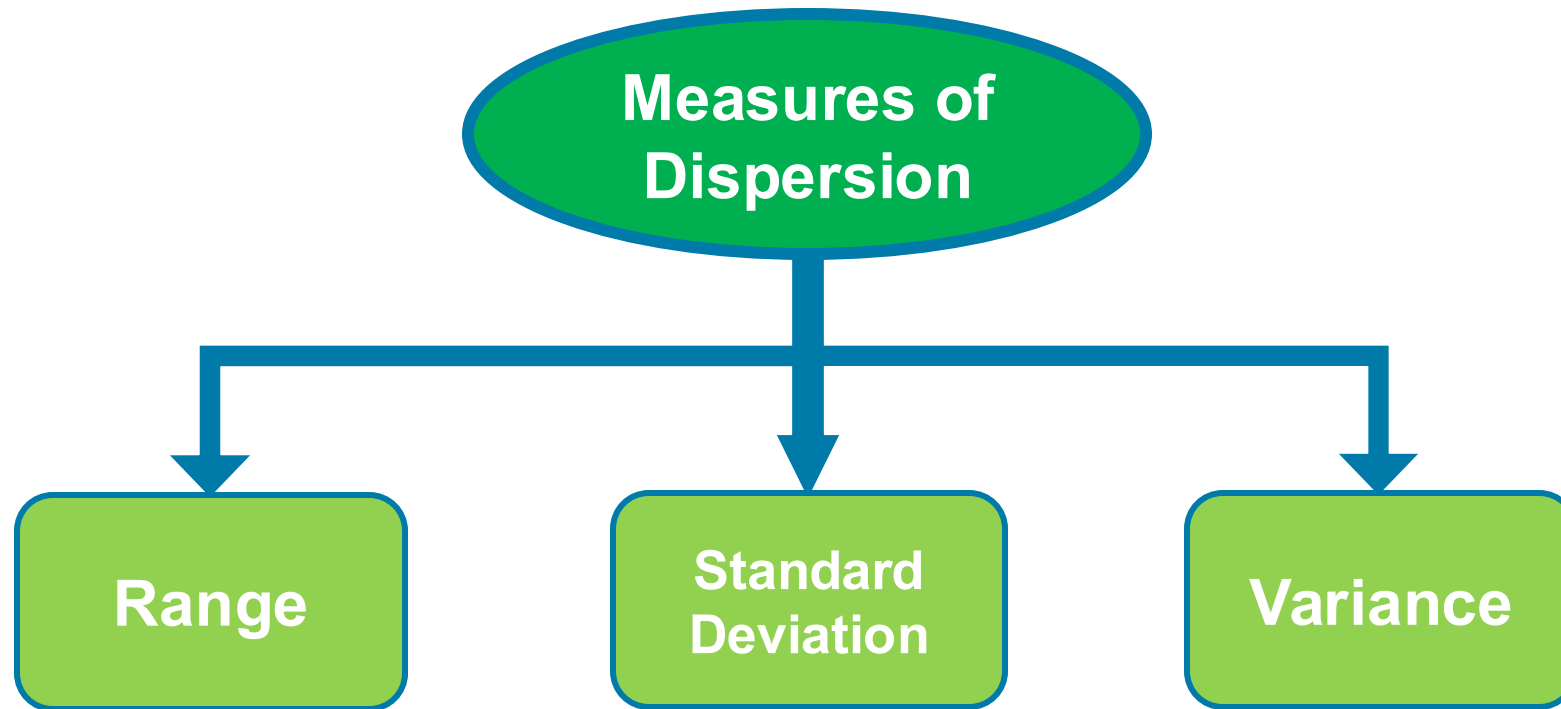


- We learned that **Measures of central tendency** (Mean, Median, Mode) describes the **central point of distribution** .
- Variability or **Measures of dispersion** describes how the scores are **scattered around the central point**.





To describe the difference in the two data sets, we use a descriptive measure that indicates the amount of **spread**, or **dispersion** in a data set.



What are Variance, Standard Deviation and Range?

- **Variance:** measures the average *squared deviation* of each data point from the mean. It shows how spread out the data is.
- **Standard Deviation:** is the *square root of variance* and gives us a more interpretable measure of spread in the same units as the original data.
- **Range:** is a measure of the spread or dispersion of a dataset. It is calculated by subtracting the *minimum* value from the *maximum* value in the dataset.

Why are they Important?

- Variance and standard deviation help you understand how much single data points vary from the mean.
- They are crucial in business for analyzing performance, risk, and consistency across data sets.





Example 1.

The data for 5 employees from **two sales teams** are given below:

Team A: 72, 73, 76, 76, 78

Team B: 67, 72, 76, 76, 84

What do you notice about
the **Mean Median and
Mode?**



It is the same for both groups
Mean = 75, Median = 76, Mode = 76

**How can we describe the difference in the two data sets, when their
Mean, Median and Mode are the same ?**



The Range

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Going back to **Example 1**:

Team A: 72, 73, 76, 76, 78

$$\text{Range for Team A} = 78 - 72 = 6$$

Team B: 67, 72, 76, 76, 84

$$\text{Range for Team B} = 84 - 67 = 17$$

- ❖ The data scores in Team A has a **smaller range** than Team B.
- ❖ The data scores in Team A are closer to each other (**more consistent**), than in Team B.



Example 2:

The following data represents the weekly earning of the Ahmed family in AED:

1000, 140, 113, 164, 90

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 1000 - 90 = \text{AED } 910\end{aligned}$$

1000, and 90 are called
extreme or outlier values

When data value is too far away from the rest of values it is called **extreme value**.

We can conclude that the **Range** is not a good measure of dispersion if we have extreme values.



Properties of the Range:

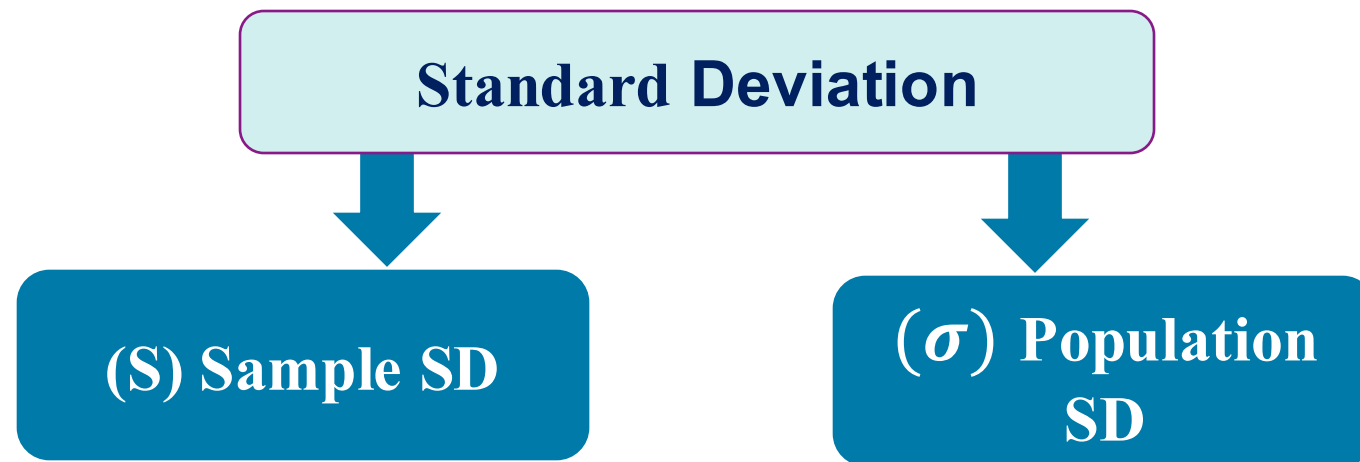
- Simple to understand.
- Quick to complete
- Not a very satisfactory measures of dispersion.
- Depends on two values only.
- Tell us nothing about the values located between the extreme ones.
- Not a good representative of the spread of data if there are extreme data.



Standard Deviation – Ungrouped data



- The Standard deviation is the number that measures how far away each number in a set of data is from their mean.
- We use the symbol ' σ ' to represent **population standard deviation** (if the data set contains all members of a specified group).
- We use the symbol '**S**' to represent **sample standard deviation** (The sample set contains part of a population).





Finding the Population Standard Deviation

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Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat

Frequency? **1** On

Step 3: Enter values. Press **=** to go the next place and right arrow  to go to next column

Step 4: Press **AC** - this stores all your values in the calculator

To clear memory and
reset calculator

SHIFT **9** **3**

For Standard Deviation of population

SHIFT **1** **4** Var

Select **3** σx

Press **=** This gives you SD

Press **x^2** **=** This gives you Variance



Find the Standard deviation,
Variance, and range of the
following set of numbers.

{7, 2, 3, 7, 8, 4, 5, 2, 7, 6, 8, 3}



Finding the Sample Standard Deviation

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Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat

Frequency? **1** On

Step 3: Enter values. Press **=** to go the next place and right arrow  to go to next column

Step 4: Press **AC** - this stores all your values in the calculator

For Standard Deviation of sample

SHIFT **1** **4** Var

Select **4** sx

Press **=** This gives you SD

Press **x^2** **=** This gives you Variance

To clear memory and reset calculator
SHIFT **9** **3**





Example:

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The achievement of two teams A and B whose scores for 10 months are given below:

Team A: 5, 9, 4, 15, 17, 30, 42, 9, 36, 5 **Team B:** 19, 35, 0, 4, 22, 2, 19, 29, 30, 3

Find the Sample Standard Deviation for Team **A** and **B**? Which Team is more consistent?

Answer:

Sample Standard Deviation **Team A:** **$S = 13.918$**

Sample Standard Deviation **Team B:** **$S = 13.098$**

Team B is **more consistent**, since it has **less Standard Deviation**.

Team A has **more variation** because it has a **higher value of Standard Deviation**.



Variance



The variance is another measure of Dispersion.

The variance is the value of the Standard Deviation **Squared**.

The sample variance is represented by s^2

The population variance is represented by σ^2

Standard deviation = $\sqrt{\text{variance}}$

Variance = $(\text{standard Deviation})^2$

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

Example 1: if the value of **Variance is 25**, then the **Standard Deviation is = 5**

Example 2: if the value of **Standard Deviation is 6**, then the **Variance is = 36**



Practice: Grouped Data



The following data shows the ages of leaders when they began their leadership

Age	Frequency
42-45	2
46-49	6
50-53	7
54-57	16
58-61	5
62-65	4
66-69	2

Find Range, Standard Deviation and Variance?



Calculating Range, Standard Deviation and Variance

Age	Frequency	Midpoint
42-45	2	43.5
46-49	6	47.5
50-53	7	51.5
54-57	16	55.5
58-61	5	59.5
62-65	4	63.5
66-69	2	67.5

$$\begin{aligned}\text{Range} &= \text{High} - \text{Low} \\ &= 69 - 42 = 27\end{aligned}$$

$$\begin{aligned}\text{Population Standard} \\ \text{Deviation} &= 5.76\end{aligned}$$

$$\begin{aligned}\text{Variance} &= 5.76^2 \\ &= 33.197\end{aligned}$$

Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat

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For Standard Deviation of population

SHIFT **1** **4** Var

Select **3** σx

Press **=** This gives you SD

Press **x^2** **=** This gives you Variance

To clear memory and reset calculator
SHIFT **9** **3**



Summary

- **Measures of dispersion** measure how spread out a data set is.
- The **more similar** the scores are to each other, the **lower** the measure of dispersion will be.
- The **less similar** the scores are to each other, the **higher** the measure of dispersion will be.
- In general, the **more spread** out a distribution is, the **larger** the measure of dispersion will be.
- The three most common measures of dispersion are: The **Range**, **Standard Deviation**, and the **Variance**.

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